Chapter-2

← Variable and datatypes →

A variable is the name given to a memory location in a program. For Example

a = 30
b = "Happy"
c = 71.22

Variable – Container to stor a value
Keywords – Reserved words in Python
Identifiers – Class/function/variable name

Data Types

Primarily there are following data types in Python

1> Integers
2> Floating point numbers
3> strings
4> Booleans
5> None

Python is a fantastic language that automatically identifies the type of data for us.

a = 71 ⇒ Identifies a as class <int>
b = 88.44 ⇒ Identifies b as class <float>
name = "Harry" ⇒ Identifies name as class <str>

Ruls for defining a variable name

→ A variable name can contain alphabets, digits and underscores.

→ A variable name can only start with alphabet and underscore.

→ A variable name can start with a digit.

→ No white space is allowed to be used in a variable name

Examples of a few variable names are:

Jay, one8, Seven_, _Seven etc

Operators in Python

Following are some common operators in Python

1> Arithmetic operators => +, -, *, / etc

2> Assignment operators => =, +=, -=, etc

3> Comparison operators => ==, >, >=, <, != etc

4> Logical operators => AND, OR, NOT etc

Typ() function and Typecasting

Type function is used to find the data type of a given variable in Python

```
a = 31
type(a) => class <int>

b = "31"
typ(b) => class <str>
```

A number can be converted into a string and vice versa (if possible).

There are many functions to convert on data type into another

```
str(31)   => "31"   => integer to string conversion
int("32") => 32     => string to integer conversion
float(32) => 32.0   => Integer to float conversion

a = 10
b = 5.5
result = a+b = 15.5 (Float)
" Integer automatically converted to float"
```

Input function

This function allows the user to take input from the Keyboard as a string.

a = input("Enter name") ⇒

Not:- ~~It~~ The output of input is alwas a string (Even if the number is entered)

Chapter 2 PracticSet

1. Write a Python program to add two numbers.
2. Write a Python program to find remainder When a number is (divisible) divided by 2.
3. Check the type of the variable assigned using input() function.
4. Use comparison operators to find out whe the a given variable 'a' is greater 'b' or not. Take a = 34 and b = 80.
5. Write a Python Programm to find average of two numbers entered by the user.
6. Write a python program to calculate square of a number entered by the user.

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EXPLORER

CHAPTER-2

- main.py
- practiceset2.ipynb

main.py

main.py > ...

```python
# variables and Datatype
name="virendra" #string
age=24 # integer
a=2.2 # Float
b=False # Boolean
e=None # none
```

File Edit Selection View Go Run ··· Chapter-2

EXPLORER

CHAPTER-2

- 1.png
- main.py
- practiceset2.ipynb

practiceset2.ipynb

practiceset2.ipynb > #Q-1 write a python program to add two numbers

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```python
# when a number is divided by 2
n=31
print(n%2)
```

[4] Python

```
1
```

```python
# Q-3 check the type of the variable assigned
# using input() function
roll_number=input()
type(roll_number)
```

Python

```
str
```

```python
# Q-4 use comparison operators to find out whether a
# given variable a is grater than b or not take a=34
# and b=80
a=input()
b=input()
a==b
```

[8] Python

```
True
```